

Layup: Orbit Fitting at LSST Scale

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Software

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Summary

The NSF-DOE Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST) is under way ([Bianco et al., 2022](#); [Ivezić et al., 2019](#); [LSST Science Collaboration et al., 2009](#)). The LSST is expected to raise the number of known solar system objects in the Minor Planet Center's catalogs to roughly 127,000 near-Earth objects (NEOs), 5.1 million main-belt asteroids (MBAs), 109,000 Jupiter Trojans, and 37,000 trans-Neptunian objects (TNOs) — four to nine times the number presently known in each class — along with 1200–2000 Centaurs, a seven- to twelve-fold increase ([Kurlander et al., 2025](#); [Murtagh et al., 2025](#)). We present Layup, an open-source package for orbit determination at LSST scale that serves as a companion to the SORCHA survey simulator ([Holman et al., 2025](#); [Merritt et al., 2025](#)). Layup is built on REBOUND ([Rein & Liu, 2012](#)) and ASSIST ([Holman et al., 2023](#)) for ephemeris-quality numerical integrations, with a C++ engine and a Python command-line interface and API. Every Layup fit reports a full state covariance, which it propagates through element and frame conversions and through ephemeris predictions to support attribution and linking.

Statement of need

Fitting orbits for the LSST data set is essential to LSST solar system science. Discovery and orbital classification are the top priorities in the Rubin Observatory LSST Solar System Science Collaboration's (SSSC's) Roadmap ([Schwamb et al., 2019](#)), but there is no orbit fitting package that can support the needs of the planetary community in the Rubin era.

44 The Minor Planet Center (MPC) fits orbits using all available observations of the object
45 reported to the MPC. Detailed population studies require orbits fit from solely LSST data
46 provided at data release (DR), and many key software utilities currently being developed
47 from the SSSC's Software Roadmap (Schwamb et al., 2019) assume an orbit fit has already
48 been generated using LSST only data. No public orbit fitting code is suitable for fitting
49 DR-only data, and the MPC software is not public. Additionally, some of the most exciting
50 science from Rubin involves the results from shifting and stacking numerous exposures with
51 KBMOD (Kernel-Based Moving Object Detection) (Smotherman et al., 2021; Whidden et
52 al., 2019), YOSO (You Only Stack Once) (Pandey et al., 2026), or heliostack (Napier et al.,
53 2026). Their detections pair a position (RA/Dec) with rates, and no orbit fitting routine takes
54 that combination as its primary input, so people synthesize tracklets from the shift-and-stack
55 sources instead — an extra processing step that can introduce correlated astrometric errors.
56 The Layup orbit fitting package fulfills all these needs.

57 State of the field

58 Three widely used open-source packages exist — Find_orb (Gray, 2022), OpenOrb (Granvik
59 et al., 2009), and OrbFit (OrbFit Consortium, 2011) — but each has limitations. None is
60 designed as a Python-native library for LSST-scale batch processing. Neither OrbFit nor
61 OpenOrb matched JPL Horizons (Giorgini et al., 1996) in detailed comparisons (Chernyavskaya
62 et al., 2021), and none handles the bound-to-unbound transition, i.e., interstellar objects
63 (Chernyavskaya et al., 2021). A more recent open-source package, GRSS (Gauss-Radau
64 Small-body Simulator) (Makadia et al., 2025), provides small-body propagation and orbit
65 determination in Python with a C++ core, but is oriented toward planetary defense — high-
66 fidelity trajectories and impact monitoring for individual objects — rather than the LSST-scale
67 survey processing that Layup targets.

68 Software design

69 The Layup orbit fitting package is built on the ASSIST small body integration package (Holman
70 et al., 2023), which itself uses REBOUND's framework (Rein & Liu, 2012) and its IAS15
71 integrator (Rein & Spiegel, 2015). ASSIST includes the terms needed for ephemeris-quality
72 accuracy — planetary and major-asteroid perturbations, general-relativistic corrections, and
73 solar oblateness. Orbit fitting minimizes the chi-square between observed and predicted
74 sky-plane positions; ASSIST supplies the partial derivatives the minimization requires.

75 Layup can ingest and fit optical astrometry, shift-and-stack observations, radar range and
76 Doppler (two-leg light time) measurements, and observations from space-based platforms.
77 Fitting begins with initial orbit determination (IOD), which produces a preliminary orbit from
78 a short arc. Layup includes three IOD methods: Gauss's method, a Bernstein-Khushalani
79 linear fit optimized for distant (outer solar system) objects, and Herget's method; Gauss
80 and Bernstein-Khushalani are selected automatically. Its modular design lets additional IOD
81 methods be incorporated easily, and the user can select a specific method or try all the available
82 options.

83 Starting from this initial estimate, Layup differentially corrects the orbit with a full least-squares
84 fit to all of the observations, in either of two parameterizations: a barycentric, equatorial,
85 Cartesian state, or a Bernstein-Khushalani basis (Bernstein & Khushalani, 2000), distance-
86 scaled parameters in a local tangent-plane reference frame. Both have six parameters and
87 share the same C++ integration and observation-modeling framework. Layup also supports
88 incremental (sequential) orbit determination, which incorporates new observations into existing
89 solutions, reproducing the accuracy of full fits in an order of magnitude less time.

90 The fits can include terms for non-gravitational accelerations, via the Marsden A1/A2/A3
91 model (Marsden et al., 1973). The radial dependence can be configured to span both asteroidal

92 and cometary laws. The amplitudes can also be fit per apparition.

93 Every fit returns a full 6×6 covariance, propagated consistently through ephemeris predictions,
94 using variational particles in REBOUND/ASSIST. We use JAX-based automatic differentiation
95 (Bradbury et al., 2018) for the Jacobians of the orbital-element and frame conversions. This
96 enables rigorous uncertainty ellipses.

97 We cross-validate Layup against JPL Horizons across MBAs, TNOs, interstellar objects and
98 radar-observed NEOs; the solutions agree to within the fit uncertainties. We further demonstrate
99 Layup's throughput by re-fitting the full MPC catalog of over 1.5 million objects, recovering
100 the MPC orbits that warm-started the fits to a median of about one part in 10^8 .

101 Although the primary goal of Layup is orbit fitting, the package also contains a set of tools
102 (orbital element conversion, ephemeris prediction, orbit visualization, and estimation of the
103 inverse of the original semimajor axis for long period comets) behind one interface. These
104 matter for following up discoveries whose observations have not yet reached the MPC, and at
105 the LSST discovery rate: existing tools handle a handful of objects rather than hundreds of
106 thousands, the JPL Horizons web API one object at a time. Ephemeris predictions with Layup
107 are highly efficient, using ASSIST's ability to integrate once and interpolate to many epochs
108 and observatory locations. All of Layup's utilities, including orbit fitting, have multiprocessing
109 built in, designed explicitly for use on a laptop or on a high-performance computing (HPC)
110 cluster.

111 Research impact statement

112 Before Layup, no open-source software supported orbit fitting at the scale of the Minor Planet
113 Center or the LSST. Layup was used for the orbit fitting of 3I/ATLAS in Rubin Observatory
114 observations of that interstellar comet (Chandler et al., 2026) — precisely the bound-to-
115 unbound case that none of the existing packages handles. Two of us (M.J.H. and K.J.N.)
116 are using Layup to fit the orbits of Rubin-discovered outer solar system objects from Rubin
117 astrometry alone. Layup is not yet embedded in a survey pipeline; that is the goal.

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AI Usage Disclosure

Portions of the Layup software and this manuscript were prepared with the assistance of OpenAI's ChatGPT (GPT-4, via the web interface; (OpenAI, 2026)) and Anthropic's Claude Opus models (including Claude Opus 5; (Anthropic, 2026b)), accessed through the Claude Code command-line assistant (Anthropic, 2026a). In the software, AI assistance was used for code conversion (Rust to C++), implementation, refactoring, test scaffolding, debugging and code review. In this paper, AI assistance was used for reference verification, copy-editing and proofreading, for organizing the manuscript into the sections above, and for drafting prose from results and decisions supplied by the authors; the Research impact statement contains the largest proportion of such text. All AI-assisted outputs were reviewed, edited and validated by the human authors — via the test suite, continuous integration and cross-validation against JPL Horizons — who made all core design and scientific decisions. H. R. did not make use of generative AI. The authors take full responsibility for the accuracy, originality, licensing, and integrity of the software and this manuscript.

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